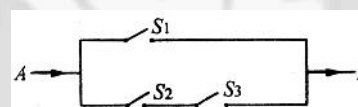


Zero Level

Probability -02

- Let E_1 and E_2 be two events such that $P(E_1) = \frac{4}{7}$ and $P(E_2) = \frac{1}{4}$, find,
 - $P(E_1 \text{ or } E_2)$, if E_1 and E_2 are mutually exclusive events.
 - $P(E_1 \text{ and } E_2)$, if E_1 and E_2 are independent events.
- Let E_1 and E_2 be events such that $P(E_1) = 0.3$, $P(E_1 \cup E_2) = 0.4$ and $P(E_2) = x$. Find the value of x such that
 - E_1 and E_2 are mutually exclusive.
 - E_1 and E_2 are independent.
- If E_1 and E_2 are independent events such that $P(E_1) = 0.3$ and $P(E_2) = 0.4$, find
 - $P(E_1 \text{ and } E_2)$
 - $P(E_1 \text{ or } E_2)$
 - $P(\bar{E}_1 \text{ and } E_2)$
 - $P(E_1 \text{ and } \bar{E}_2)$
 - $P(\bar{E}_1 \text{ and } \bar{E}_2)$
- If E and F are two events such that $P(E) = \frac{1}{4}$, $P(F) = \frac{1}{3}$ and $P(E \cap F) = \frac{1}{2}$, show that E and F are independent events.
- Let E and F be events such that $P(E) = \frac{1}{2}$, $P(F) = \frac{7}{12}$ and $P(\text{not } E \text{ or not } F) = \frac{1}{4}$. State whether E and F are :
 - mutually exclusive,
 - independent.
- Amit and Nisha appear for an interview for two vacancies in a company. The probability of Amit's selection is $\frac{1}{5}$ and that of Nisha's selection is $\frac{1}{6}$. What is the probability that :
 - both of them are selected ?
 - only one of them is selected ?
 - one of them is selected ?
- Kamal and Vimal appeared for an interview for two vacancies. The probability of Kamal's selection is $\frac{1}{3}$ and that of Vimal's selection is $\frac{1}{5}$. Find the probability that only one of them will be selected.
- Arun and Ved appeared for an interview for two vacancies. The probability of Arun's selection is $\frac{1}{4}$ and that of Ved's rejection is $\frac{2}{3}$. Find the probability that at least one of them will be selected.
- A can solve 90% of the problems given in a book, and B can solve 70%. What is the probability that at least one of them will solve a problem selected at random from the book ?
- Given the probability that A can solve a problem is $\frac{2}{3}$, and the probability that B can solve the same problem is $\frac{3}{5}$, find the probability that

- At least one of A and B will solve the problem,
 - One of the two will solve the problem.
- The probabilities of a specific problem being solved independently by A and B are $\frac{1}{2}$ and $\frac{1}{3}$ respectively. If both try to solve the problem independently, find the probability that
 - the problem is solved
 - exactly one of them solve the problem.
 - A problem is given to three students whose chances of solving it are $\frac{1}{4}, \frac{1}{5}, \frac{1}{6}$ respectively. Find the probability that the problem is solved.
 - The probabilities of A, B, C solving a problem are $\frac{1}{3}, \frac{1}{4}$ and $\frac{1}{6}$ respectively. If all the three try to solve the problem simultaneously, find the probability that exactly one of them will solve it.
 - A, B, C shoot to hit a target. If A hits the target 4 times in 5 trials, B hits it 3 times in 4 trials and C hits it 2 times in 3 trials, what is the probability that the target is hit by at least 2 persons ?
 - An anti-aircraft gun can take a maximum of 4 shots of an enemy plane moving away from it. The probabilities of hitting the plane of the first, second, third and fourth shots are 0.4, 0.3, 0.2 and 0.1 respectively. What is the probability that at least one shot hits the plane ?
 - Two persons A and B throw a coin alternatively till one of them gets a 'head' and wins the game. Find their respective probabilities of winning if A starts first.
 - Three persons A, B, C throw a die in succession till one of them gets a 'six' and wins the game. Find their respective probabilities of winning.
 - A coin is tossed. If a head comes up, a die is thrown but if a tail comes up, the coin is tossed again. Find the probabilities of obtaining
 - two tails
 - a head and the number 6
 - a head and an even number.
 - The probability that a certain team wins is 0.7, that it loses is 0.2 and that it has a tie is 0.1. The team plays three games. Find the probability that the team wins at least two of the games but does not lose any.
 - An electric system has three switches S_1, S_2 and S_3 as shown below : The switches operate independent of the one another, and the current will flow from A to B either when S_1 is closed or when S_2 and S_3 are closed. If their probabilities of working be given by $P(S_1) = P(S_2) = P(S_3) = \frac{1}{2}$, find the probability that the current flows from A to B.



21. A box contains 3 white and 5 red balls. Two balls are drawn at random, one at a time, without replacement. Find the probability of getting 1 white and 1 red ball.
22. A bag contains 7 white, 4 red and 9 black balls. If four balls are drawn one by one without replacement, what is the probability that all are white ?
23. A bag contains 4 white, 7 black and 5 red balls. Three balls are drawn one after the other without replacement. Find the probability that the balls drawn are white, black and red respectively.
24. A bag contains 8 red and 6 green balls. Three balls are drawn one by one without replacement. Find the probability that at least two balls drawn are green.
25. Two cards are drawn one by one without replacement from a pack of 52 cards. Find the probability that both are kings.
26. Two cards are drawn one by one without replacement from a well-shuffled pack of 52 cards. Find the probability that neither a jack nor a card of spade is drawn.
27. Suppose a sample of 100 electric bulbs contains 5 defective bulbs. Two electric bulbs are randomly selected one after the other without replacement. If either bulb is defective, the entire lot is rejected. Find the probability that the sample will be rejected.
28. A bag contains 4 white and 2 black balls. Another bag contains 3 white and 5 black balls. If one ball is drawn from each bag, find the probability that
 - (i) both are white,
 - (ii) both are black,
 - (iii) one is white and the other is black.
29. Bag A contains 4 red and 5 black balls and bag B contains 3 red and 7 black balls. One ball is drawn from bag A and two from bag B. Find the probability that out of three balls drawn, two are black and one is red.
30. Three bags contain 5 white and 6 red; 8 white and 4 red; 6 white and 4 red balls respectively. One ball is drawn at random from each bag. Find the probability that all of them are of the same colour.
31. A purse contains 2 silver and 4 copper coins. A second purse contains 4 silver and 3 copper coins. If a coin is pulled out at random from one of the two purses, what is the probability that it is silver coin ?
32. A bag contains 4 red and 3 black balls. A second bag contains 2 red and 4 black balls. One bag is selected at random. From the selected bag, one ball is drawn. Find the probability that the ball drawn is red.
33. One bag contains 5 white and 6 black balls. Another bag contains 7 white and 3 black balls. One ball at random is transferred from the first bag to the second bag and then a ball is drawn from the second bag. Find the probability that the ball drawn is white.
34. A bag contains 4 yellow and 5 red balls and another bag contains 6 yellow and 3 red balls. A ball is drawn from the first bag and without seeing its colour, it is put into the second bag. Now, a ball is drawn from the second bag. Find the probability that the ball drawn is yellow in colour.
35. A class consists of 10 boys and 8 girls. Three students are selected at random. What is the probability that the selected group has
 - (i) all boys ?
 - (ii) all girls ?
 - (iii) 1 boy and 2 girls ?
 - (iv) at least one girl ?
 - (v) at most one girl ?
36. In a group of students, there are 3 boys and 3 girls. Four students are to be selected at random from the group. Find the probability that either 3 boys and 1 girl; or 3 girls and 1 boy are selected.
37. From a bag containing 20 tickets, numbered from 1 to 20, two tickets are drawn at random. Find the probability that
 - (i) both the tickets have prime numbers on them,
 - (ii) on one there is a prime number and on the other there is multiple of 4.
38. A bag contains 8 red, 3 white and 9 blue balls. If three balls are drawn at random, find the probability that
 - (i) all the three balls are blue balls,
 - (ii) all the balls are of different colours.
39. An urn contains 7 white, 5 black and 3 red balls. Two balls are drawn at random. Find the probability that
 - (i) both the balls are red,
 - (ii) one ball is red and other is black,
 - (iii) one ball is white.
40. A bag contains 30 tickets, numbered from 1 to 30. Five tickets are drawn at random and arranged in ascending order. Find the probability that the third number is 20.
41. A box contains 4 defective and 9 non-defective bulbs. Two bulbs are chosen at random. What is the probability that both the bulbs are non-defective ?
42. Three cards are drawn at random from a pack of well-shuffled 52 cards. Find the probability that :
 - (i) all the three cards are of the same suit,
 - (ii) one is a king, the other is a queen and the third is a jack.
43. Find the probability of drawing a diamond card in each of two consecutive draws from a well shuffled pack of cards, if
 - (i) the card drawn is not replaced after the first draw,
 - (ii) the card drawn is replaced after the first draw.
44. A word has 9 letters, consisting of 5 consonants and 4 vowels. Three letters are chosen at random. What is the probability that more than 1 vowel will be selected ?
45. A letter of the English alphabet is chosen at random. Calculate the probability that the letter so chosen
 - (i) is a vowel
 - (ii) precedes K and is a vowel,
 - (iii) follows K and is a vowel.
46. Find the probability that when a hand of seven cards is dealt from a well-shuffled deck of 52 cards, it contains :
 - (i) all 4 kings,
 - (ii) exactly 3 kings,
 - (iii) at least 3 kings.
47. Four cards are drawn at random from a pack of 52 playing cards. Find the probability of getting :

- (i) all the four cards of the same suit,
(ii) all the four cards of the same number,
(iii) one card from each suit,
(iv) two red cards and two black cards,
(v) all cards of the same colour,
(vi) all face cards.
48. Two cards are drawn from a well-shuffled deck of 52 cards. Find the probability of getting exactly one ace.
49. In a certain region of Russia, the probability that a person lives at least 80 years is 0.75 and the probability that he or she lives at least 90 years is 0.63. What is the probability that a randomly selected 80 years old person from this region will survive to become 90 ?
50. From the set all families with two children, a family is selected at random and is found to have a girl. What is the probability that the other child of the family is a girl ? Assume that in a two-child family all sex distributions are equally probable.
51. Find the probability that a single toss a die results in a number less than 4 if it is given that the toss resulted in an odd number.
52. We draw eight cards at random from an ordinary deck of 52 cards. Given that three of them are spades. What is the probability that the remaining five are also spades ?
53. A green and red die are rolled :
(a) Find the conditional probability of obtaining a sum greater than 10, given that the red die resulted in a 5.
(b) Find the conditional probability of obtaining sum 7, given that the red die resulted in a number less than 4.
54. Three indistinguishable objects are distributed in three cells. Find the conditional probability that all three occupy the same cell, given that at least two of them are in the same cell.
55. Urn I contains three green and five red balls. Urn II contains two green, one red and two yellow balls. We select an urn at random and then draw one ball at random from that urn. What is the probability that we obtain a green ball ?
56. From an ordinary deck of 52 cards, cards are drawn one by one, at random and without replacement. What is the probability the fourth heart is drawn on the tenth draw ?
57. Two cards are drawn from a well-shuffled ordinary deck of 52 cards. Find the probability that they are both aces if the first card is (a) replaced, (b) not replaced.
58. Box I contains 3 red and 2 blue marbles while box II contains 2 red and 8 blue marbles. A fair coin is tossed. If the coin turns up head a marble is chosen from box I, if it turns up tail a marble is chosen from box II. Find the probability that a red marble is chosen.
59. Suppose that five good and two defective fuses have been mixed up. To find the defective ones, we test them one by one, at random, and without replacement. What is the probability that we find both the defective fuses is exactly three tests ?
60. If 5% of men and 0.25% of women are colour blind, what is the probability that a randomly selected person is colour blind ?

61. Two cards from an ordinary deck of 52 cards are missing. What is the probability that a random card drawn from this deck is a spade ?
62. Box I contains 2 red and 5 blue marbles while box II contains 3 red and 7 blue marbles. A fair coin is tossed. Box I, if it turns up head a marble is chosen from box II. Find the probability that a blue marble is chosen.
63. Box I contains 2 red and 5 white balls, while box II contains 4 red and 2 white balls. A ball is chosen at random from the first box and placed in the second box without observing its colour. Then a ball is drawn from the second box. Find the probability that it is white.
64. Urn A contains 6 red and 4 black balls and urn B contains 4 red and 6 black balls. One ball is drawn at random from urn A and placed in urn B. Then one ball is drawn at random from urn B and placed in urn A. If one ball is now drawn at random from urn A. Find the probability that it is found to be red.
65. A tells truth with-probability p_1 and B with probability p_2 . If they make the same statement. Find the probability that their statement is true.
66. A man is known to speak the truth 3 times out of 4 times. He throws a die and reports that it is six. Find the probability that it is actually a six.
67. Urn I contains 4 white and 6 black balls. Urn II contains 5 white and 5 black balls. One of two urns is chosen at random and two balls are drawn from it without replacement. Both the balls turn out to be black. What is the probability that it came from the first urn ?
68. In Wellville, 55% of the voters are registered Republicans and 45% are registered Democrats. There are two candidates for mayor : R , a Republican, and D , a Democrat. In the election, 80% of the Republicans and 10% of the Democrats voted for R . 20% of the Republicans and 90% of the Democrats voter for D . If a voter is selected at random and is found to have voted for D . What is the probability that he is a Democrat ?
69. Suppose three urns A, B and C contains white, red and black balls as given below :

Urn	White	Red	Black
A	10	5	10
B	8	8	4
C	5	5	5

- Two balls are drawn at random and found to be white and black. What is the probability that they came from urn C ?
70. Urn I has 2 white and 3 black balls, Urn II, 4 white and 1 black; Urn III, 3 white and 4 black. An urn is selected at random and a ball is drawn at random is found to be white. Find the probability that urn I was selected.
71. In a bolt factory, there are four machines A, B, C and D manufacturing 20%, 15%, 25% and 40% of the total output respectively. Of their outputs 5%, 4%, 3% and 2% in the same order are defective bolts. A bolt is chosen at random from the factory's production and is found defective. What is the probability that the bolt was manufactured by machine A or machine D ?

72. A random variable X has the following probability distribution :

X	0	1	2	3
$P(X)$	0.3	k	$2k$	0.4

Find the value of k .

73. Find the probability distribution of number of tails when two coins are tossed simultaneously.
74. Find the probability distribution of number of tails in the simultaneous toss of three coins and also find the mean and variance of number of tails.
75. Find the probability distribution of number of heads in the simultaneous toss of four coins and also find the mean and variance of number of heads.
76. Find the probability distribution of number of successes in two tosses of a die where a success is defined as 'a number greater than 4'.
77. A die is tossed thrice. A success is getting a 2 or a 5 or an 8. Find the probability distribution of the number of success and also find the mean and variance of successes.
78. Find the probability distribution of the random variable X which denotes the number of times "a total of 9" appears in two throws of a pair of dice.
79. Three cards are drawn successively with replacement from a well-shuffled deck of 52 cards. A random variable X denotes the number of hearts in the three cards drawn. Determine probability distribution of X and also find the mean and variance of hearts.
80. Two cards are drawn successively with replacement from a well-shuffled pack of 52 cards. Find the probability distribution of the number of kings, also find the mean and variance of kings.
81. Two cards are drawn without replacement from a well-shuffled deck of cards. Determine the probability distribution of the number of 'face cards.'
82. Two cards are drawn one by one without replacement from a well-shuffled pack of 52 cards. Find the probability distribution of the number of kings.
83. From a lot of 30 bulbs which include 6 defective one, a sample of 4 bulbs is drawn at random with replacement. Find the probability distribution of the number of defective bulbs, also find the mean of defective bulbs.
84. Three defective bulbs are mixed with 7 good ones. Find the probability distribution of the number of defective bulbs, if 3 bulbs are drawn at random.
85. Four rotten oranges are accidentally mixed with 16 good ones. Three oranges are drawn at random from the mixed lot. Find the probability distribution of X , the number of rotten oranges.
86. Four bad eggs are mixed with 10 good ones. If 3 eggs are drawn one by one without replacement, find the probability distribution of the number of the bad eggs drawn.
87. Three cards are drawn successively with replacement from a well-shuffled deck of 52 cards. A random variable X denotes the number of spades in three cards. Determine the probability distribution of X also find the mean of X and variance of X .

88. A coin is tossed until the head appears, or the tail appears 4 times in succession. Find the probability distribution of the number of tosses also find the mean of tosses.

89. Let X be a random variable which assumes the values x_1, x_2, x_3, x_4 such that

$$2P(X = x_1) = 3P(X = x_2) = P(X = x_3) = 5P(X = x_4)$$

Find the probability distribution of X .

90. State whether it will be possible for a random variable to have any of following probability distributions. Give reasons :

(i)

X	0	1	2
$P(X)$	0.3	0.5	0.2

(ii)

X	0	1	2
$P(X)$	0.6	0.2	0.1

(iii)

X	0	1	2	3	4
$P(X)$	0.2	0.4	0.3	-0.1	0.2

91. A random variable X has the following probability distribution :

X	0	1	2	3	4	5	6	7
$P(X)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Find the value of :

(i) k (ii) $P(X < 3)$ (iii) $P(X \geq 6)$ (iv) $P(0 < x < 3)$

92. Find the probability that in tossing a fair coin 6 times there will appear

- (a) 0 (b) 1 (c) 2 (d) 3
(e) 4 (f) 5 (g) 6 heads

93. What is the probability that 3 tails appear when an unbiased coin is tossed 5 times ?

94. If, in general, 30% of patients afflicted with a certain disease die from it, what is the probability that just a two die in a group of five ?

95. What is the probability of obtaining the sum 7 just twice in four rolls of a pair of dice ?

96. In five throws with an ordinary dice what is the chance of throwing three aces at least ?

97. If on an average 1 vessel in every 10 is wrecked, what is the probability that out of 5 vessels expected to arrive, 4 at least will arrive safely.

98. Find the probability that in five tosses of a fair die a 3 appears (a) twice, (b) at most once, (c) at least two times.

99. Find the probability that in a family of 4 children there will be (a) at least 1 boy, (b) at least 1 boy and at least 1 girl. Assume that the probability of male birth is $1/2$.

100. Find the probability of getting a total of 7 at least once in three tosses of a pair of a fair dice.

101. If 20% of the bolts produced by a machine are defective, determine the probability that out of 4 bolts chosen at random (a) 1, (b) 0, (c) less than 2, bolts will be defective.

102. Find the probability of getting a total of 11 (i) once, (ii) twice, in two tosses of a pair of fair dice.

103. What is the probability of getting a 9 exactly one in 3 throws with a pair of dice ?



PROBABILITY-02(ANSWERS)

1. (i) $\frac{23}{28}$ (ii) $\frac{1}{7}$
2. (i) $x = 0.1$ (ii) $x = 1/7$
3. (i) 0.12 (ii) 0.58 (iii) 0.28 (iv) 0.18 (v) 0.42
5. (i) No (ii) No
6. (i) $1/30$ (ii) $3/10$ (iii) $2/3$
7. $2/5$ 8. $1/2$ 9. 0.97 10. (i) $13/15$ (ii) $2/15$
11. (i) $2/3$ (ii) $1/2$ 12. $1/2$ 13. 72 14. $5/6$
15. 0.6976 16. $P(A \text{ wins}) = 2/3$, $P(B \text{ wins}) = 1/3$
17. $P(A \text{ wins}) = 36/91$, $P(B \text{ wins}) = 30/91$, $P(C \text{ wins}) = 25/91$
18. (i) $1/8$ (ii) $1/8$ (iii) $3/8$
19. 0.147 20. $5/8$ 21. $15/28$ 22. $7/969$ 23. $1/24$
24. $5/13$ 25. $1/221$ 26. $105/221$ 27. $97/990$
28. (i) $1/4$ (ii) $5/24$ (iii) $13/24$
29. $7/15$ 30. $26/99$ 31. $91/42$ 32. $19/42$ 33. $82/121$
34. $29/45$
35. (i) $5/34$ (ii) $7/102$ (iii) $3/102$ (iv) $29/34$ (v) $10/17$
36. $2/5$ 37. (i) $14/95$ (ii) $4/19$ 38. (i) $7/95$ (ii) $18/95$
39. (i) $1/35$ (ii) $1/7$ (iii) $8/15$ 40. $285/5278$
41. $6/13$ 42. (i) $44/85$ (ii) $3/5525$ 43. (i) $1/17$ (ii) $1/16$
44. $17/42$ 45. (i) $5/26$ (ii) $3/26$ (iii) $1/13$
46. (i) $1/7735$ (ii) $9/1547$ (iii) $46/7735$
47. (i) $44/4165$ (ii) $1/20825$ (iii) $2197/20825$
- (iv) $169/435$ (v) $1196/10829$ (vi) $4/592$
48. $32/221$ 49. 0.84 50. $1/3$ 51. $2/3$
52. 0.00000544 53. (a) $1/6$ (b) $1/6$
54. $1/7$ 55. $31/80$ 56. 0.059 57. (a) $1/169$ (b) $1/221$
58. $2/5$ 59. $2/21$ 60. 0.02625 61. $1/4$
62. $99/140$ 63. $21/56$ 64. $23/55$
65. $\frac{1}{2}[p_1 p_2 + (1-p_1)(1-p_2)]$ 66. $3/8$ 67. $6/11$ 68. 0.786
69. 0.325 70. $14/57$ 71. 0.3157, 0.254
72. $k = 0.1$
90. (i) Yes, since $\sum P(X) = 1$ and each $P(X) > 0$
- (ii) No, since $\sum P(X) \neq 1$
- (iii) No, since $P(3) < 0$
91. (i) $k = 1/10$ (ii) $3/8$ (iii) $19/100$ (iv) $3/10$
92. (a) $1/64$ (b) $3/32$ (c) $15/64$ (d) $5/16$ (e) $15/64$
- (f) $3/32$ (g) $1/64$
93. $5/16$
94. $3087/10000$
95. $25/216$
96. $23/648$
97. $45297/50000$
98. (a) $625/3888$ (b) $3125/3888$ (c) $763/3888$
99. (a) $15/16$ (b) $7/8$
100. $91/216$
101. (a) $256/625$ (b) $256/625$ (c) $512/625$
102. (i) $17/162$ (ii) $1/324$
103. $64/243$

